

Franco Terranova

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EDUCATION

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| Université de Lorraine & INRIA, Nancy, France                                                                                                                                                                                                                                                                                                                                                                                                                            | 10/2023 — 10/2026                |
| Ph.D. in Computer Science - RESIST Team / SuperviZ Project (France 2030)                                                                                                                                                                                                                                                                                                                                                                                                 |                                  |
| <ul style="list-style-type: none"><li>• Design of a sequential decision-making framework for identifying critical vulnerability paths in networks</li><li>• Design of invariant and transferable embedding action spaces for scalable and generalizable RL solutions</li><li>• Extension of Microsoft CyberBattleSim with a more realistic cyber-attack simulation</li><li>• Developed a LM-based framework for vulnerability analysis and dynamics estimation</li></ul> |                                  |
| University of Pisa, Pisa, Italy                                                                                                                                                                                                                                                                                                                                                                                                                                          | 09/2021 — 09/2023                |
| Master's Degree in Artificial Intelligence and Data Engineering                                                                                                                                                                                                                                                                                                                                                                                                          | Final Grade: (110/110) cum laude |
| University of Pisa, Pisa, Italy                                                                                                                                                                                                                                                                                                                                                                                                                                          | 09/2018 — 07/2021                |
| Bachelor's Degree in Computer Engineering                                                                                                                                                                                                                                                                                                                                                                                                                                | Final Grade: (110/110) cum laude |

EXPERIENCE

|                                                                                                                   |                   |
|-------------------------------------------------------------------------------------------------------------------|-------------------|
| Universitat Politècnica de Catalunya (UPC)                                                                        | Barcelona, Spain  |
| Visiting Researcher — Barcelona Neural Networking Center                                                          | 11/2025 – 03/2026 |
| Project: GNN-based latent action embedding spaces for enhancing RL heuristics in graph combinatorial optimization |                   |
| Collaboration with the Geometric Intelligence Lab, UC Santa Barbara                                               |                   |
| University of Waterloo                                                                                            | Waterloo, Canada  |
| Visiting Researcher — Cheriton School of Computer Science                                                         | 06/2024 — 12/2024 |
| Project: Multi-agent RL for a secure and efficient virtual machine placement                                      |                   |
| Fermi National Accelerator Laboratory                                                                             | Chicago, Illinois |
| Master Degree Thesis Researcher — Deep Skies Lab                                                                  | 05/2023 — 08/2023 |
| Project: Deep RL for autonomous telescope scheduling                                                              |                   |
| European Space Agency                                                                                             | Köln, Germany     |
| AI/ML Engineer Intern — Spaceship EAC Team                                                                        | 11/2022 - 04/2023 |
| Project: Deep learning for real-time detection of spaceflight-related ocular conditions                           |                   |
| Fermi National Accelerator Laboratory                                                                             | Chicago, Illinois |
| Summer Intern — GlidenWMS Project                                                                                 | 07/2022 — 09/2022 |
| Project: Flexible pilot jobs framework for distributed high throughput computing                                  |                   |

ACADEMIC ENGAGEMENT

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|--------------------------------------------------------------------------|---------------------------------------------------------|
| Teaching Assistant                                                       |                                                         |
| Data Science                                                             | IDMC, Nancy, France - 03/2026                           |
| Introduction to Natural Language Processing                              | IDMC, Nancy, France - 09/2025                           |
| Project Management Tools                                                 | IDMC, Nancy, France - 09/2025                           |
| Probability and Statistics                                               | IDMC, Nancy, France - 09/2025                           |
| Data Science                                                             | IDMC, Nancy, France - 03/2025                           |
| Méthodes et Outils pour l'Analyse du Comportement Humain                 | IDMC, Nancy, France - 01/2025                           |
| Symbolic Artificial Intelligence                                         | IDMC, Nancy, France - 01/2025                           |
| Tutorials                                                                |                                                         |
| Tutorial From RL to Meta-RL                                              | European Agent Systems Summer School, Romania - 09/2025 |
| Introduction to Reinforcement Learning & Deep RL                         | DeepLorIA, France - 11/2024                             |
| Foundations of Deep RL and Environment Setup                             | European Agent Systems Summer School, Ireland - 08/2024 |
| Organizer of the DeepLorIA network: delivered 10+ tutorials on AI topics | LORIA, Nancy, France - 2024–2026                        |
| Talks & Seminars                                                         |                                                         |
| Learning Graph-RL Heuristics for Automated Attack Path Planning          | CISPA, Saarbrücken, Germany - 06/2026                   |

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|-----------------------------------------------------------------------------|-------------------------------------------------------------|
| Learning Generalizable Graph Heuristics via RL and Applications             | EPFL, Switzerland - 06/2026                                 |
| A More Realistic Simulation for AI-driven Attack Path Discovery             | Black Hat Asia 2026, Singapore - 04/2026                    |
| Sélection de Paradigmes et Validation Statistique: Bonnes Pratiques en ML   | University of Lorraine, France - 12/2025                    |
| Modeling Attacker Behavior with RL Agents to Anticipate Vulnerability Paths | University of Tokyo, Japan - 11/2025                        |
| Learning to Predict Cyber Attack Paths with Reinforcement Learning          | Cybersecurity Workshop, Tokyo, Japan - 11/2025              |
| Can AI Help Us Understand Vulnerabilities?                                  | Forum InCyber Europe, Lille, France - 03/2025               |
| Discovering Critical Vulnerability Paths using RL Agents                    | Campus Cyber, Paris, France - 03/2025                       |
| Deep RL for Cyber-Attack Path Prediction                                    | INRIA Rennes, France - 03/2024                              |
| Autonomous Telescope Scheduling with Reinforcement Learning                 | PyHEP 2023 Workshop, Online - 10/2023                       |
| Introduction to Reinforcement Learning                                      | Fermilab Cross-lab AI Meetings, Chicago, Illinois - 07/2023 |

### Summer Schools

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|-----------------------------------------------|-----------------------------|
| OxML Summer School on Representation Learning | Oxford, UK - 08/2025        |
| Geilo Winter School on Graphs & Applications  | Geilo, Norway - 01/2024     |
| Fermilab Summer School                        | Chicago, Illinois - 07/2022 |

### Released Tools

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| <i>LaGCO-RL</i> : An RL Framework for Graph Combinatorial Optimization                             | Github - 05/2026 |
| <i>CVE-LMTune</i> : Tuning LMs for Multi-Taxonomy Vulnerability Classification                     | Github - 01/2026 |
| <i>Continuous CyberBattleSim</i> : A More Realistic Simulation for RL-driven Attack Path Discovery | Github - 07/2025 |

**Peer Reviewing:** IEEE TNSM, NeurIPS 2026, IJCNN 2026, NOMS 2026, NeurIPS 2025, IJCNN 2025, RESSI 2025, AAAI 2024 (Workshop), Computational and Structural Biotechnology Journal 2024

**Supervision:** 1 IDMC M2 Master Thesis, 1 IDMC M1 Summer Intern, 4 IDMC M1 Semester Students & 1 University of Cincinnati Intern on the project “Automated Analysis of Vulnerabilities Using NLP” (2025)

### EXTRA

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**Patent:** Ritter, S., Drescher, J., Stern, C., Terranova, F., Cowley, A. (2025). “Mobile Device and Computer-Implemented Method for Real-Time Retinal Diagnosis, Data Processing Apparatus, Computer Program, and Computer-Readable Medium,” filed in US and EU patent offices.

**Mentoring:** Mentee for the Université de Lorraine Mentorship Program (2024), LeadTheFuture Mentorship Program (2023), Young ISSNAF Mentoring Program for Students (2023), and SGAC Mentorship Program (2023)

**Memberships:** Member of the AI Doctoral Academy (since 2023), IEEE, IEEE Computer Society, IEEE Computational Intelligence Society & IEEE Communications Society Member (since 2024), ACM, and ACM SIGAI Member (since 2024)

**Social Engagement:** AI-enabled Proactive Cybersecurity Stand, Nuit de la Science & Fête de la Science, Nancy, France (2025); Scrutineer for the European Elections, Nancy, France (2024); Booth Staff for the European Space Agency at Festival Yggdrasil, Eurexpo Lyon, France (2023)

### OTHER PROJECTS

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|----------------------------------------------------------------------------------------|-------------------|
| <b>Internet of Things for a Smart Regolith Transportation System</b>                   | Golden, Colorado  |
| <i>Over The Dusty Moon Challenge 2023</i> - Colorado School of Mines / Lockheed Martin | 12/2022 — 06/2023 |
| <b>Cyber-Attack Response with Graph Neural Networks and Language Models</b>            | Milan, Italy      |
| <i>European Union Defense Innovation Scheme Hackaton 2024</i>                          | 05/2024 — 06/2024 |

### LANGUAGES

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**Italian:** Native    **English:** C1/C2 (Professional proficiency)    **French:** C1 (Advanced)    **Spanish:** B1 (Intermediate)